Independent Audit of the Two-Body Toda Package

Assessment of `phase2/twobody_toda.tex` and `phase2/scripts/twobody_toda_*.py`

Independent Audit

2026

Abstract

We report an independent mathematical and numerical audit of the two-body Toda Bethe-ansatz package `phase2/twobody_toda.tex` (which inputs `phase2/twobody_toda_numbers.tex`) and its four verification scripts `phase2/scripts/twobody_toda_*.py`. Every analytical identity is re-derived symbolically and tested in 70-digit arithmetic using an independent Fejér first-rule (Chebyshev-root) Fredholm solver, 45-digit de Hoog Laplace inversion, and Neville/Padé extrapolators in `phase2/scripts/audit_twobody_toda.py`.

1. **Closed form of $\vartheta_N(r)$ and relation to Phase 2h/2i (Points 7 & 9):** We **confirm (c)** that in the iterated edge-then-semiclassical limit ($h \to 0$ at fixed y_a, then $Y \to \infty$ at fixed $r = \sqrt{y_2/y_1} \in (0,1)$), the antisymmetrised two-term directional residual equals

$$\vartheta_N^{(1)}(r) = \pi - \frac{4}{\pi}\chi_2(r) + \frac{2}{\pi}\left(r - \frac{1}{r}\right), \qquad \chi_2(r) = \sum_{m=0}^{\infty} \frac{r^{2m+1}}{(2m+1)^2}.$$

   We prove that the Phase 2h/2i residual $R_0(\infty, r) = -\frac{4}{\pi}(\chi_2(r) - r)$ confirmed in `phase2/audit_phase2i.tex` is the *forward* (unsymmetrised) two-term limit $R_{12}^{(1)}(r)$. Subtracting the non-vanishing symmetric limit $S^{(1)}(r) = \frac{2}{\pi}(r + r^{-1}) - \pi$ dictated by exact two-term reciprocity $H(u,v) + H(v,u) = 2B(u)B(v) - 2\pi$ accounts for all three structural differences: $+\pi$ arises from the endpoint contact constant $-B(0^+)^2 = -\pi$, $-\frac{2}{\pi}r^{-1}$ from the one-loop Jacobian dressing of the slower soliton $y_2 = r^2 Y$, and $+\frac{2}{\pi}r$ from $+\frac{4}{\pi}r$ minus the Jacobian dressing $+\frac{2}{\pi}r$ of the faster soliton $y_1 = Y$.

2. **High-priority audit of the four-term Krein source in `audit_frame_dictionary.tex` (Point 8):** We **confirm (c)** that `phase2/audit_frame_dictionary.tex` (Eq. (9), line 431) and `audit_frame_dictionary.py` (line 797) printed an incorrect characteristic source $2f(u)[B(v) - \sqrt{\pi}]$ instead of the true source $(\partial_u + \partial_v)\widehat{H}(u,v) = 2f(u)B(v) - 2\sqrt{\pi}f(v)$; the error arose from applying ∂_u instead of ∂_v to the one-body edge reference $U(v) = 2\sqrt{\pi}[B(v) - \sqrt{\pi}]$. However, we prove by three independent arguments (homogeneous characteristic uniqueness at fixed one-body spectrum, exact vanishing of all 3×3 Schmidt determinants and 2×2 mixed-partial determinants of the true source versus non-zero determinants for $(\partial_{y_1} + \partial_{y_2})A_1(y_1, y_2)$, and spatial cluster separability) that **the verdict of `audit_frame_dictionary.tex` excluding the antisymmetric two-body freedom $A_\lambda(y_1, y_2)$ on the Toda side survives unchanged**.

3. **Head-on directional coefficient and error budgets (Points 2 & 10):** We **confirm (c)** that the defined two-term head-on observable $R_{12}^{\mathrm{ho}}(y_1, y_2; h)$ vanishes at $O(h^0)$, but **qualify (d)** its physical interpretation because the algebraically antisymmetric endpoint representative $\mathcal{B}_{\mathrm{ho}}/h \to (B(u_1) - \sqrt{\pi})(B(u_2) - \sqrt{\pi})$ does not vanish after subtracting $4rY$. We also **qualify** the error budgets at large Y in the h-fit ($\kappa_1 = 1.453 > 1$ at $Y = 64, h = 0.02$) and at small $r = 0.25$ in the $1/Y$-fit ($y_2 = r^2 Y = 2$ at $Y = 32$, causing a 2684× truncation-error jump in $S^{(1)}(0.25)$ that can be eliminated by using the exact one-body symmetric limit).

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1 Summary of Verdicts

Claims are labelled according to `AGENTS.md`: **(a)** proved in the literature, **(b)** conjectured or heuristic in the literature, **(c)** derived in this project, and **(d)** a guess or convention choice. Each claim is assigned **Confirmed**, **Qualified**, or **Refuted**.

ID / Ref	Claim in `twobody_toda.tex`	Label	Verdict	Independent audit evidence
C1 Eqs. (1)–(3), §1	Classical/one-loop Toda-to-good-Boussinesq dictionary ($b = \mathrm{e}^{-1/2}$, $\kappa_a = \sqrt{hy_a/b}$, sound shift $\delta c_s = -\frac{3}{4\pi}h$) and overtaking/head-on $h \to 0$ limits of $(\delta_T + C_{12})/h$.	(a)/(c)	**Confirmed** (minor test gap)	Symbolic series in `audit_twobody_toda.py` confirms $(\delta_T + C_{12})/h \to H_{\mathrm{cl}}(Y, r)$ (overtaking) and $0 + O(h)$ (head-on). Package script tested only δ_T/h, omitting C_{12}/h.
C2 Eqs. (7)–(11), §2.2	Finite-sea endpoint decomposition $F_{12} = \widetilde{F}_{12} - \frac{D_1^-}{\mathcal{D}(q)}\xi(p_2)$ and exact antisymmetric representative $\mathcal{B}_{12} = F_{12} - M(p_1)Q(p_2) = -\mathcal{B}_{21}$ before $h \to 0$.	(c)	**Confirmed**	Verified in 70-digit arithmetic on an independent Fejér-I Chebyshev mesh ($N = 44, 64$, $h = 0.45$, $q = 1.25$): max algebraic error 5.66×10^{-73}, Fredholm integration-by-parts error 7.52×10^{-20}.
C3 Eqs. (12)–(13), §3.1	Edge-limit Wiener–Hopf reduction $N_{12}/h \to H(u_1, u_2)$, $U_{12}/h \to 2\sqrt{\pi}[B(u_2) - \sqrt{\pi}]$, and two-term reciprocity $H(u, v) + H(v, u) = 2B(u)B(v) - 2\pi$.	(c)	**Confirmed**	Confirmed by 5-point Fejér-I finite-sea Neville extrapolation ($h \in [0.004, 0.064]$) matching 26-digit de Hoog WH convolution to $< 5 \times 10^{-7}$ for R_{12} and $< 5 \times 10^{-6}$ for ϑ_N and $\mathcal{B}_{12}/h$.

ID / Ref	Claim in `twobody_toda.tex`	Label	Verdict	Independent audit evidence
C4 Eq. (14), §3.1 **(Point 8)**	Four-term edge observable $\widehat{H}(u,v) = H(u,v) - 2\sqrt{\pi}[B(v) - \sqrt{\pi}]$ satisfies $(\partial_u + \partial_v)\widehat{H} = 2f(u)B(v) - 2\sqrt{\pi}f(v)$; `audit_frame_dictionary.tex` printed an incorrect source $2f(u)[B(v) - \sqrt{\pi}]$.	(c)	**Confirmed** (prior exclusion verdict survives)	Symbolic & 45-digit de Hoog check ($< 10^{-18}$) confirms discrepancy $2\sqrt{\pi}[f(u) - f(v)]$. However, homogeneous uniqueness, $\det_{3\times3} S_{\text{true}} = 0$, and $\det_{2\times2} \partial_u \partial_v S_{\text{true}} = 0$ prove exclusion of $A_\lambda(y_1, y_2)$ in `audit_frame_dictionary.tex` survives unchanged (§3).
C5 Eqs. (16)–(18), §3.2 **(Points 7, 9)**	Regularised representation Eq. (16) and closed form $\vartheta_N^{(1)}(r) = \pi - \frac{4}{\pi}\chi_2(r) + \frac{2}{\pi}(r - r^{-1})$ from $R_{12}^{(1)}(r) = -\frac{4}{\pi}(\chi_2(r) - r)$ and $S^{(1)}(r) = \frac{2}{\pi}(r + r^{-1}) - \pi$.	(c)	**Confirmed** (Eq. 16 untested in pkg)	Derived analytically via integration by parts and verified by independent de Hoog/Fejér WH solver across $r \in [0.25, 0.9]$. Direct check of untested Eq. (16) passes to 7.6×10^{-15}. Explains exact relation to $R_0(\infty, r)$ of Phase 2h/2i (§4).
C6 Eq. (19), §3.3 **(Point 10)**	Leading head-on directional coefficient vanishes: $R_{12}^{\text{ho},(0)}(y_1, y_2) = 0$, with $N_{12}^{\text{ho}}/h = O(h)$, $U_{12}^{\text{ho}}/h = O(h)$, and $(\delta_T^{\text{ho}} + C_{12}^{\text{ho}})/h = O(h)$.	(c)/(d)	**Confirmed** (c) **Qualified** (d)	Confirmed for R_{12}^{ho} ($< 2 \times 10^{-6}$ in 5-point Fejér-I Neville extrapolation). Qualified as a physical scattering phase: $\mathcal{B}_{\text{ho}}/h \to (B(u_1) - \sqrt{\pi})(B(u_2) - \sqrt{\pi})$ has non-zero $O(\sqrt{Y})$ and $O(1)$ residuals after subtracting $4rY$ (§6).
C7 §2.1, §4, Tables 1–2 **(Point 2)**	Error budgets separate quadrature mesh error, $h \to 0$ fit spread, and $Y \to \infty$ extrapolation uncertainty across all grid points.	(c)	**Qualified**	At $Y = 64$, `quartic7` includes $h = 0.02$ where $\kappa_1 = 1.453 > 1$, inflating h-spread 3800×; `hlog7` ℓ^1 norm 37.23 inflates `mesh_error`. At $r = 0.25$, $y_2 = r^2 Y = 2$ at $Y = 32$ causes an 8.52×10^{-5} error in $S^{(1)}$ (2.76× larger than `fit_spread`), curable to 3.18×10^{-8} using exact 1-body $S^{(1)}$ (§5).

2 Audit Methodology, Script Execution, and Conventions (Points 1, 3, 4, 5, 6)

2.1 Execution and Runtimes of the Package Scripts (Point 6)

To audit the four package scripts without violating the repository rule against modifying existing files, `phase2/scripts/audit_twobody_toda.py` executes each script via `runpy.run_path` inside a read-only `Path.write_text` guard. Without this guard, running `twobody_toda_identities.py`, `twobody_toda_edge.py`, or `twobody_toda_analysis.py` directly overwrites tracked repository files (`twobody_toda_identities.json`, `twobody_toda_edge.json`, `twobody_toda_grid.json`, and `twobody_toda_numbers.tex`) by updating runtime metadata or regenerating tables. Inside the guard, we verify that every script prints `ALL CHECKS PASSED`, that the payloads generated in memory for `twobody_toda_grid.json` and `twobody_toda_numbers.tex` match the tracked files byte-for-byte, and that the SHA-256 hashes of all ten package files remain unchanged.

Table 2: Execution status, cached runtimes measured by `audit_twobody_toda.py`, and full cache-generation runtimes recorded in the package JSON artifacts.

Package script	Output status	Cached (s)	Cache gen. (s)	Side effect if run unguarde
`twobody_toda_identities.py`	`ALL CHECKS PASSED`	38.0–52.0	56.45	Rewrites `identities.json`
`twobody_toda_scan.py`	`ALL CHECKS PASSED`	0.01	953.37	Read-only if 14 rows present
`twobody_toda_edge.py`	`ALL CHECKS PASSED`	52.55	1135.79	Rewrites `edge.json`
`twobody_toda_analysis.py`	`ALL CHECKS PASSED`	9.99	9.99	Rewrites `grid.json`, `numbers`

2.2 Conventions and Independence from Earlier Refuted Claims (Points 3 & 5)

Good Boussinesq and Toda scaling conventions (Point 3). The package works in the fixed project convention (`AGENTS.md`) for the **good** Boussinesq equation $u_{tt} = u_{xx} - u_{xxxx} - 3(u^2)_{xx}$ with $u = \varphi_x$, $L = \frac{1}{2}\varphi_t^2 - \frac{1}{2}\varphi_x^2 + \varphi_x^3 - \frac{1}{2}\varphi_{xx}^2$, and $\Omega = \int \mathrm{d}\pi \wedge \mathrm{d}\varphi$. In the semiclassical Toda dictionary (`phase2/audit_toda_dictionary.tex`, `phase2/audit_phase2i.tex`), the edge parameter is $b = \mathrm{e}^{-1/2}$, the rapidities are $\kappa_a = \sqrt{hy_a/b}$, and the rescaled energies and momenta are $E_{\mathrm{good}}(\kappa_a) = \frac{2}{3}b(\sinh 3\kappa_a - 3\sinh\kappa_a)$, $P_{\mathrm{good}}(\kappa_a) = 4b(\kappa_a\cosh\kappa_a - \sinh\kappa_a)$, and $M_{\mathrm{good}}(\kappa_a) = 2b\sinh\kappa_a$ **(a)**/**(c)**. The one-loop Toda sound speed $c_s = 1 - \frac{3}{4\pi}h + O(h^2)$ matches our independent 5-point Fejér-I Neville extrapolation $(c_s - 1)/h \to -0.23873241498$, agreeing with $-\frac{3}{4\pi} = -0.23873241464$ to 3.4×10^{-10} **(c)**.

Regarding the defect scale convention (`AGENTS.md`: scattering uses $s = \sigma_{\mathrm{vac}}$, whereas the spectrum uses $\eta\sigma_{\mathrm{vac}}$; see `phase2/audit_smx.tex` and `phase2/errata.tex` Item E1), `phase2/twobody_toda.tex` operates entirely within the Toda Bethe ansatz and its Wiener–Hopf edge limit and never invokes the Boussinesq integrable defect transmission matrix or the Corrigan–Zambon (`arXiv:2306.03772`, verified against repository dictionary) defect potential. Its results are therefore unaffected by the defect s convention **(c)**.

Reliance on earlier audited results (Point 5). We cross-checked `phase2/twobody_toda.tex` against all prior audit reports (`audit_phase2i.tex`, `audit_frame_dictionary.tex`, `audit_toda_dictionary.t` `audit_onebody_scheme.tex`, `audit_smx.tex`, and `errata.tex`). The package does not rely on any refuted claim:

- It explicitly respects the negative verdict of `audit_phase2i.tex` that the naive directional Bethe phase N_{12} violates reciprocity by $N_{12} + N_{21} = O(h)$, and separates the symmetric part $S^{(1)}(r)$ from the antisymmetric part $\vartheta_N^{(1)}(r)$.
- It respects the verdict of `audit_frame_dictionary.tex` that no first-principles Boussinesq-to-Toda two-body frame dictionary has been derived, framing its main result strictly as a conditional sharp target (Section 5).
- Its dispute with the printed Krein source formula in `audit_frame_dictionary.tex` is examined in detail in Section 3.

2.3 Audit of Claim Coverage and Circularity (Points 1 & 4)

Every **(c)** claim in `phase2/twobody_toda.tex` was checked against the four package scripts for coverage and non-circularity:

1. **Untested regularised integral representation Eq. (16) (`eq:regular`):** In Section 3.2, the package derives the closed form of $R_{12}^{(1)}(r)$ by changing variables $t = u(z)$ in the Wiener–Hopf convolution $H(u,v) = 2\int_0^v f(u - v + t)B(t)\,\mathrm{d}t$ using $B(u(z))u'(z) = 2\sqrt{z}$ to obtain

$$H(u(Y), u(r^2Y)) = 4\int_0^{r^2Y} \sqrt{z}\, f(u(Y) - u(r^2Y) + u(z))\,\mathrm{d}z. \tag{1}$$

However, none of the four package scripts tests Eq. (1): `twobody_toda_edge.py` integrates exclusively in $t \in [0, v]$. In Check 4 of `audit_twobody_toda.py`, we test Eq. (1) directly at $(Y, r) = (4, 0.5)$ via $z = s^2$ on a 10-node Fejér-I mesh, confirming agreement with the t-integral to 7.60×10^{-15} **(c)**.

2. **Incomplete check of $(\delta_T + C_{12})/h$ in `twobody_toda_identities.py`:** Lines 113–120 of `twobody_toda_identities.py` check only δ_T/h at $h = 10^{-30}$ ($2Y(1-r^2)\log[(1-r)/(1+r)]$ for overtaking and $4rY$ for head-on), omitting the classical frame subtraction $C_{12}/h = -\kappa_1(P_{\text{good}}(\kappa_2) - M_{\text{good}}(\kappa_2))/h$, which contributes $+4rY$ in the overtaking channel (producing the $+4rY$ term in $H_{\text{cl}}(Y, r)$, Eq. 15) and $-4rY$ in the head-on channel (cancelling $\delta_T^{\text{ho}}/h \to +4rY$ to $O(h)$). Check 2 of `audit_twobody_toda.py` verifies the complete symbolic Laurent expansion of $(\delta_T + C_{12})/h$ in both channels **(c)**.

3. **Non-circularity (Point 4):** Neither the finite-sea polynomial extrapolations (`quartic7` in h) nor the Wiener–Hopf semiclassical extrapolations (`quad3` in $1/Y$) feed the closed-form target $\vartheta_N^{(1)}(r)$ back into the fits. In `twobody_toda_edge.py`, the analytic coefficients $A = 1/(2\pi)$ and $d_0 = \frac{1}{48} - \frac{1}{4\pi^2}$ are used only as the initial guess for Newton's method when solving $\Phi(u) = \frac{4}{3}y^{3/2}$ to 10^{-26} tolerance, which does not bias the converged root. However, in `twobody_toda_analysis.py` (line 228), the assertion threshold `5e-4` on the semiclassical intercept was chosen large enough to pass the 8.52×10^{-5} truncation error at $r = 0.25$ (analysed in Section 5).

3 High-Priority Item: The Four-Term Krein Source and the Verdict of `audit_frame_dictionary.tex` (Point 8)

3.1 Exact Origin of the Discrepancy in the Krein Characteristic Source

In the edge limit $p_a = q + hu_a$ ($u_1 = u > u_2 = v > 0$), the directional Bethe phase N_{12}/h and endpoint reference U_{12}/h converge to

$$H(u, v) = 2\int_0^v f(u - v + t)B(t)\,\mathrm{d}t, \qquad U(v) = 2c\big[B(v) - c\big], \qquad c = B(0^+) = \sqrt{\pi}, \tag{2}$$

where $B(u) = \Phi'(u)$ and $f(u) = B'(u) = \Phi''(u)$ **(c)**. Differentiating $H(u, v)$ along the characteristic direction $\partial_u + \partial_v$ (along which $u - v$ is constant) gives purely the upper-limit boundary contribution:

$$(\partial_u + \partial_v)H(u, v) = \frac{\mathrm{d}}{\mathrm{d}\epsilon}\bigg|_{\epsilon=0} 2\int_0^{v+\epsilon} f(u - v + t)B(t)\,\mathrm{d}t = 2f(u)B(v) \qquad (u > v > 0). \tag{3}$$

Crucially, the one-body endpoint subtraction $U(v) = 2c[B(v) - c]$ depends *only on the second argument* v: $\partial_u U(v) = 0$ and $\partial_v U(v) = 2cB'(v) = 2cf(v)$. Therefore, the edge limit $\widehat{H}(u, v) = H(u, v) - U(v)$ of the four-term subtraction $F_{12} = N_{12} - U_{12}$ satisfies **(c)**

$$(\partial_u + \partial_v)\widehat{H}(u, v) = 2f(u)B(v) - 2cf(v) = 2f(u)B(v) - 2\sqrt{\pi}f(v), \tag{4}$$

which is Eq. (14) of `phase2/twobody_toda.tex`.

By contrast, `phase2/audit_frame_dictionary.tex` (Eq. (9), line 431) and `phase2/scripts/audit_frame_c` (line 797) wrote

$$(\partial_{u_1} + \partial_{u_2})F_{\text{sub}}(u_1, u_2) \overset{\text{printed}}{=} 2f(u_1)\big[B(u_2) - \sqrt{\pi}\big] = 2f(u_1)B(u_2) - 2\sqrt{\pi}f(u_1). \tag{5}$$

Equation (5) committed the error of replacing $\partial_{u_2}B(u_2) = f(u_2)$ with $f(u_1)$ in the subtracted term, as if $2\sqrt{\pi}c$ had been subtracted inside the t-integrand of $H(u_1, u_2)$ at the upper limit $t = u_2$ rather than subtracted as a function of u_2 alone. The difference between the true source (4) and the printed formula (5) is

$$S_{\text{true}}(u, v) - S_{\text{printed}}(u, v) = 2\sqrt{\pi}\big[f(u) - f(v)\big] \neq 0 \qquad (u \neq v). \tag{6}$$

In Check 3 of `audit_twobody_toda.py`, we verify Eq. (4) both symbolically and numerically to 45-digit precision using de Hoog Laplace inversion and 4th-order central differences along $u+\epsilon, v+\epsilon$: at $(u,v)=(4,1)$, $(\partial_u+\partial_v)\widehat{H} = -0.347104254662920455 35$, matching $S_{\text{true}}(4,1)$ to 2.06×10^{-21} while disagreeing with $S_{\text{printed}}(4,1) = +0.83774806069206012686$ by 1.1848523153549805822 **(c)**.

3.2 Why the Verdict of `audit_frame_dictionary.tex` Excluding $A_\lambda(y_1,y_2)$ Survives Unchanged

In `phase2/frame_dictionary.tex`, a one-parameter family of classical antisymmetric two-body observables

$$A_\lambda(y_1,y_2) = \lambda\,\frac{(y_1-y_2)(y_1y_2)^{3/2}}{y_1^2+y_2^2} \tag{7}$$

was shown to preserve antisymmetry, scale invariance, and the overtaking classical limit $H_{\text{cl}}(Y,r)$. In `phase2/audit_frame_dictionary.tex`, Theorem 3.2 concluded that cluster separability and the Krein characteristic PDE exclude $A_\lambda(y_1,y_2)$ $(\lambda\neq0)$ on the Toda side. Because `audit_frame_dictionary.te` cited the rank-1 factorisation of the printed source (5), we must determine whether Theorem 3.2 survives when S_{printed} is replaced by the true source $S_{\text{true}}(u,v) = 2f(u)B(v) - 2cf(v)$. We prove that **the exclusion verdict survives completely**, for three independent reasons **(c)**:

1. **Homogeneous characteristic uniqueness at fixed one-body spectrum:** The functions $B(u)=\Phi'(u)$ and $f(u)=B'(u)$ are one-body spectral quantities determined solely by the single-hole Wiener–Hopf dressing on the half-line. Any admissible two-body modification $\widetilde{H}(u,v) = \widehat{H}(u,v) + \Delta\widehat{H}(u,v)$ that leaves the one-body dispersion relation $\Phi(u)$ (and hence $B(u)$, $f(u)$, and $S_{\text{true}}(u,v)$) unchanged must satisfy the *same* inhomogeneous Krein PDE (4). Consequently, the difference $\Delta\widehat{H}(u,v)$ must satisfy the **homogeneous** characteristic equation

$$(\partial_u+\partial_v)\Delta\widehat{H}(u,v) = 0 \qquad (u>v>0), \qquad \Delta\widehat{H}(u,0) = 0, \tag{8}$$

   whose unique solution on $u\geq v\geq 0$ is $\Delta\widehat{H}(u,v)\equiv 0$, *independently of the form of the inhomogeneous source* $S_{\text{true}}(u,v)$. In fact, lines 800–804 of `audit_frame_dictionary.py` tested $A_\lambda(y_1,y_2)$ by differentiating $A_1(y_1,y_2)$ directly without ever evaluating S_{printed} in code.

2. **Exact algebraic PDE invariants (3×3 Schmidt determinant and 2×2 mixed-partial determinant):** Although $S_{\text{true}}(u,v) = 2f(u)B(v) - 2cf(v)$ has Schmidt rank 2 rather than rank 1 (being a sum of $f(u)\otimes B(v)$ and $1\otimes f(v)$), it satisfies two exact algebraic identities that do not depend on knowing B or f in advance:

   (a) Every 3×3 determinant $\det_{1\leq i,j\leq3} S_{\text{true}}(u_i,v_j)$ vanishes identically because S_{true} lies in a 2-dimensional tensor subspace; and

   (b) Taking ∂_u annihilates the one-body subtraction $-2cf(v)$, so the mixed partial derivative

$$\partial_u\partial_v(\partial_u+\partial_v)\widehat{H}(u,v) = \partial_u\partial_v S_{\text{true}}(u,v) = 2f'(u)f(v)$$

   is *strictly rank-1 separable*, meaning every 2×2 cross-determinant vanishes identically:

$$\det_{1\leq i,j\leq2}\big[\partial_{u_i}\partial_{v_j}S_{\text{true}}(u_i,v_j)\big] \equiv 0.$$

   In Check 3 of `audit_twobody_toda.py`, we evaluate both determinants symbolically for $D_A(y_1,y_2) = (\partial_{y_1}+\partial_{y_2})A_1(y_1,y_2)$: at $(y_1)\in(1,2,3)$ and $(y_2)\in(4,5,6)$, the 3×3 Schmidt determinant is

$$\det_{1\leq i,j\leq3} D_A(y_{1,i},y_{2,j}) = -\frac{19469608614112801}{66954376636977070080000}\sqrt{30} \neq 0,$$

and at $(y_1) \in (1,2)$, $(y_2) \in (4,5)$, the 2×2 mixed-partial determinant is

$$\det_{1\le i,j\le 2} \partial_{y_1}\partial_{y_2} D_A(y_{1,i}, y_{2,j}) = \frac{8835448337}{86838197512500}\sqrt{5} \neq 0.$$

Hence $A_\lambda(y_1, y_2)$ cannot be absorbed into any rank-2 Krein source of the form $2\tilde{f}(y_1)\tilde{B}(y_2) - 2c\tilde{f}(y_2)$ for *any* one-body functions $(\tilde{B}, \tilde{f})$.

3. **Spatial cluster separability and r-analyticity:** Independently of the Krein PDE, any canonical frame transformation generated by local one-body asymptotic invariants $\sum_{a\neq b} c_k Q_k(p_a)\tilde{Q}_k(p_b)$ has finite Schmidt rank ($\det_{3\times 3} = 0$), whereas $\det_{1\le i,j\le 3} A_1(y_{1,i}, y_{2,j}) = -\frac{1054728\sqrt{30}}{291949675} \neq 0$. Moreover, $A_\lambda(Y, r^2Y) = \lambda\frac{r^3(1-r^2)}{1+r^4}Y^2$ has rational poles at $r^4 = -1$ ($r = \mathrm{e}^{\pm \mathrm{i}\pi/4}$) on the unit circle, whereas the Toda Bethe kernel has logarithmic branch singularities only at $r = \pm 1$.

4 Independent Derivation and Verification of $\vartheta_N(r)$ and Relation to Phase 2h/2i (Points 7 & 9)

4.1 Analytic Derivation from the Edge-Limit Bethe Equations (Point 7)

In the edge limit $p_a = q + hu_a$ with $u_a = u(y_a)$ fixed by $\Phi(u(y)) = \frac{4}{3}y^{3/2}$, the large-$y$ Wiener–Hopf asymptotics of $\Phi(u)$, $B(u) = \Phi'(u)$, and $f(u) = B'(u)$ follow from expanding $\widehat{\Phi}(s) = \frac{\sqrt{\pi}}{s^2}g(s)$ around $s \to 0^+$:

$$u(y) = y - A\log y + c_0 + \frac{d_0}{y} + \frac{e_0}{y^2} + O(y^{-3}), \qquad A = \frac{1}{2\pi}, \qquad d_0 = \frac{1}{48} - \frac{1}{4\pi^2}. \tag{9}$$

Since $B(u(y))u'(y) = \frac{\mathrm{d}}{\mathrm{d}y}\Phi(u(y)) = 2\sqrt{y}$, differentiating $u(y)$ yields the exact Jacobian expansion **(c)**

$$B(u(y)) = \frac{2\sqrt{y}}{u'(y)} = 2\sqrt{y}\left[1 + \frac{A}{y} + \frac{A^2 + d_0}{y^2} + \frac{A^3 + 2Ad_0 + 2e_0}{y^3} + O(y^{-4})\right], \tag{10}$$

and differentiating a second time gives $f(u) = \frac{1}{\sqrt{u}} - \frac{A}{u^{3/2}} + O(u^{-5/2})$.

Substituting $u_1 = u(Y)$, $u_2 = u(r^2Y)$, and $z = Yt$ ($t \in (0, r^2)$) into the regularised integral (1) gives

$$u(Y) - u(r^2Y) + u(Yt) = Y(1 - r^2 + t) + A\log\left(\frac{r^2}{t}\right) - A\log(1 - r^2 + t) + O(Y^{-1}).$$

Expanding $f(w) = w^{-1/2} - Aw^{-3/2} + O(w^{-5/2})$ at $w = Y(1 - r^2 + t) + \Delta$ gives

$$f\big(u(Y) - u(r^2Y) + u(Yt)\big) = \frac{1}{\sqrt{Y(1 - r^2 + t)}} - \frac{A\big[2 + \log(r^2/t)\big]}{2Y^{3/2}(1 - r^2 + t)^{3/2}} + O(Y^{-5/2}\log Y).$$

Integrating the $O(Y^{-1/2})$ term against $4Y^{3/2}\sqrt{t}\,\mathrm{d}t$ reproduces the classical overtaking benchmark

$$H_{\mathrm{cl}}(Y, r) = 4Y\int_0^{r^2}\sqrt{\frac{t}{1 - r^2 + t}}\,\mathrm{d}t = 4rY + 2Y(1 - r^2)\log\left(\frac{1 - r}{1 + r}\right),$$

while the $O(Y^{-3/2})$ term gives the exact $Y \to \infty$ forward directional residual **(c)**:

$$R_{12}^{(1)}(r) = \lim_{Y\to\infty}\big[H(u(Y), u(r^2Y)) - H_{\mathrm{cl}}(Y, r)\big] = -2A\int_0^{r^2}\frac{\sqrt{t}\,\big[2 + \log(r^2/t)\big]}{(1 - r^2 + t)^{3/2}}\,\mathrm{d}t. \tag{11}$$

To evaluate Eq. (11) in closed form, set $s = \sqrt{t/(1 - r^2 + t)} \in (0, r)$, so that $t = (1-r^2)s^2/(1-s^2)$, $1 - r^2 + t = (1 - r^2)/(1 - s^2)$, and $\frac{\sqrt{t}}{(1-r^2+t)^{3/2}}\,\mathrm{d}t = \frac{2s^2}{1-s^2}\,\mathrm{d}s$. Then $\log(r^2/t) - \log(1 - r^2 + t) = -2\log(s/r) - \log(1 - r^2)$, and using $\frac{s^2}{1-s^2} = \frac{\mathrm{d}}{\mathrm{d}s}\big[-s + \frac{1}{2}\log\frac{1+s}{1-s}\big]$ to integrate by parts (the boundary terms vanish at $s = 0$ as $O(s^3\log s)$ and at $s = r$ since $\log(r/r) = 0$) yields **(c)**:

$$R_{12}^{(1)}(r) = 8A\int_0^r \log\left(\frac{s}{r}\right)\frac{s^2}{1 - s^2}\,\mathrm{d}s = -8A\big[\chi_2(r) - r\big] = -\frac{4}{\pi}\big[\chi_2(r) - r\big]. \tag{12}$$

4.2 Exact Comparison with $R_0(\infty, r)$ of Phase 2h/2i: Origin of π and r^{-1} (Point 9)

Equation (12) establishes that the forward two-term residual $R_{12}^{(1)}(r)$ is *identically equal* to the Phase 2h/2i constant-term residual confirmed in `phase2/audit_phase2i.tex`:

$$R_{12}^{(1)}(r) \equiv R_0(\infty, r) = -\frac{4}{\pi}[\chi_2(r) - r]. \tag{13}$$

Why, then, does the antisymmetrised residual $\vartheta_N^{(1)}(r) = \frac{1}{2}(R_{12}^{(1)}(r) - R_{21}^{(1)}(r))$ equal $\pi - \frac{4}{\pi}\chi_2(r) + \frac{2}{\pi}(r - r^{-1})$ rather than $R_0(\infty, r)$?

In Phase 2h/2i, only the forward branch R_{12} ($y_1 > y_2$) was computed, under the tacit assumption that the two-term residual was antisymmetric. As proved in `phase2/audit_phase2i.tex` (Claim 3) and Section 3.1 of `twobody_toda.tex`, integration by parts on $H(u, v) = 2\int_0^v f(u - v + t)B(t)\,\mathrm{d}t$ yields the exact two-term reciprocity failure

$$H(u_1, u_2) + H(u_2, u_1) = 2B(u_1)B(u_2) - 2B(0^+)^2 = 2B(u_1)B(u_2) - 2\pi. \tag{14}$$

Meanwhile, in the classical benchmark, $\delta_T(y_1, y_2) + \delta_T(y_2, y_1) = 0$, whereas the classical frame terms sum to $(C_{12} + C_{21})/h \to 8\sqrt{y_1 y_2} = 8rY$. Therefore, the symmetric combination $S(Y, r) = \frac{1}{2}[R_{12}(Y, r) + R_{21}(Y, r)]$ is controlled purely by the one-body Jacobian $B(u(y))$ in Eq. (10):

$$\begin{aligned} S(Y, r) &= B(u(Y))B(u(r^2 Y)) - \pi - 4rY \\ &= 4rY\left(1 + \frac{A}{Y} + \frac{A^2 + d_0}{Y^2} + \cdots\right)\left(1 + \frac{A}{r^2 Y} + \frac{A^2 + d_0}{r^4 Y^2} + \cdots\right) - \pi - 4rY \\ &= \underbrace{\left[\frac{2}{\pi}\left(r + \frac{1}{r}\right) - \pi\right]}_{S^{(1)}(r)} + \left[\frac{1}{\pi^2 r} + 4\left(\frac{1}{4\pi^2} + d_0\right)\left(r + \frac{1}{r^3}\right)\right]\frac{1}{Y} + O(r^{-5}Y^{-2}). \end{aligned} \tag{15}$$

Since $\vartheta_N(Y, r) = \frac{1}{2}(R_{12} - R_{21}) = R_{12}(Y, r) - S(Y, r)$, subtracting $S^{(1)}(r)$ from $R_{12}^{(1)}(r) = R_0(\infty, r)$ gives **(c)**:

$$\vartheta_N^{(1)}(r) = R_0(\infty, r) - S^{(1)}(r) = \left[-\frac{4}{\pi}\chi_2(r) + \frac{4}{\pi}r\right] - \left[\frac{2}{\pi}r + \frac{2}{\pi}\frac{1}{r} - \pi\right] = \pi - \frac{4}{\pi}\chi_2(r) + \frac{2}{\pi}\left(r - \frac{1}{r}\right). \tag{16}$$

Equation (16) accounts for every structural difference between $\vartheta_N^{(1)}(r)$ and $R_0(\infty, r)$:

- **The $+\pi$ constant:** Arises from $-(-B(0^+)^2) = +\pi$ in the two-term reciprocity identity (14), i.e., the non-vanishing boundary density $B(0^+) = \sqrt{\pi}$ at the spectral edge.
- **The $-\frac{2}{\pi}r^{-1}$ pole as $r \to 0^+$:** Arises from the one-loop Jacobian dressing $B(u(y_2)) = 2\sqrt{y_2}[1 + \frac{1}{2\pi y_2} + \cdots]$ of the *slower* soliton $y_2 = r^2 Y$ multiplying the classical factor $2\sqrt{y_1} = 2\sqrt{Y}$ in $B(u(y_1))B(u(y_2)) - 4rY$.
- **The reduction of $+\frac{4}{\pi}r$ to $+\frac{2}{\pi}r$:** Half of the $+\frac{4}{\pi}r$ linear term in $R_0(\infty, r)$ is cancelled by the one-loop Jacobian dressing $+\frac{2}{\pi}r$ of the *faster* soliton $y_1 = Y$ multiplying $2\sqrt{y_2} = 2r\sqrt{Y}$ in $S^{(1)}(r)$.

4.3 Independent Numerical Verification of $\vartheta_N^{(1)}(r)$ (Point 7)

In Check 4 of `audit_twobody_toda.py`, we verify Eq. (16) numerically using two independent methods:

1. **Fresh de Hoog / Fejér-I Wiener–Hopf solver:** We invert $\widehat{\Phi}(s)$, $\widehat{B}(s)$, and $\widehat{f}(s)$ using the de Hoog algorithm (rather than Talbot contours) at 26-digit precision and integrate $H(u, v)$ on geometrically subdivided intervals $[0, 1/3, 1, 3, 9, \ldots, v]$ using 10-point Fejér-I Chebyshev quadrature on an independent grid $Y \in \{16, 32, 64, 128\}$ at $r \in \{0.25, 0.5, 0.75\}$.

2. **4-point Neville and Padé [2/1] extrapolations across all six $r \in \{0.25, 0.4, 0.5, 0.6, 0.75, 0.9\}$:** Furthermore, because $S^{(1)}(r) = \frac{2}{\pi}(r + r^{-1}) - \pi$ is fixed analytically by the one-body function $B(u(y))$, we compare the naive extrapolation of $\vartheta_N(Y, r) = R_{12}(Y, r) - S(Y, r)$ against the hybrid estimator $\vartheta_N^{\text{1-body-}S}(r) := \lim_{Y\to\infty} R_{12}(Y, r) - S^{(1)}(r)$, which avoids the severe small-y_2 truncation error in $S(Y, r)$ at small r.

Table 3: Independent numerical verification of $R_{12}^{(1)}(r)$, $S^{(1)}(r)$, and $\vartheta_N^{(1)}(r)$ across $r \in [0.25, 0.9]$ using 4-point Neville extrapolation and the exact one-body symmetric subtraction $S^{(1)}(r)$ (from `audit_twobody_toda.py`).

r	$R_{12}^{(1)}(r)$ (exact)	$\lvert R_{12}^{\text{Nev}} - R_{12}^{\text{exact}}\rvert$	$\vartheta_N^{(1)}(r)$ (exact Eq. 18)	$\lvert\vartheta_N^{\text{Nev}} - \vartheta_N^{\text{exact}}\rvert$ (raw)	$\lvert(R_{12}^{\text{Nev}} - S^{(1)}) - \vartheta_N^{\text{exact}}\rvert$
0.25	−0.00226186980	3.18×10^{-8}	+0.43369675123	8.52×10^{-5}	$\mathbf{3.18 \times 10^{-8}}$
0.40	−0.00962285926	1.81×10^{-12}	+1.28577245446	9.41×10^{-10}	$\mathbf{1.81 \times 10^{-12}}$
0.50	−0.01951540939	3.44×10^{-13}	+1.53052781328	1.26×10^{-10}	$\mathbf{3.44 \times 10^{-13}}$
0.60	−0.03545557168	3.13×10^{-13}	+1.66313226454	2.53×10^{-11}	$\mathbf{3.13 \times 10^{-13}}$
0.75	−0.07717783630	2.97×10^{-8}	+1.73812362486	6.07×10^{-8}	$\mathbf{2.97 \times 10^{-8}}$
0.90	−0.16035045278	4.01×10^{-7}	+1.70092910305	4.14×10^{-7}	$\mathbf{4.01 \times 10^{-7}}$

5 Audit of the Error Budgets (Point 2)

The package separates three error sources: quadrature mesh error, $h \to 0$ fit spread across alternative polynomial bases, and $Y \to \infty$ extrapolation uncertainty. Our audit confirms that the separation is structurally transparent, but identifies three quantitative weaknesses that require **qualification**:

1. **Breakdown of the small-κ regime at $Y = 64$ in the $h \to 0$ fit:** The package's primary $h \to 0$ fit (`quartic7`) uses the 7 smallest values $h \in \{0.001, 0.0015, 0.002, 0.003, 0.005, 0.01, 0.02\}$, while its comparison family includes `quintic8` ($h \le 0.03$), `quad4` ($h \le 0.003$), and `hlog7` ($h \le 0.01$ with $h^2 \log h, h^3 \log h$). However, the Boussinesq rapidity is $\kappa_1 = \sqrt{hY/b}$ with $b = \mathrm{e}^{-1/2} \approx 0.60653$. At $Y = 64$:

$$\kappa_1(Y = 64, h = 0.02) = 1.4526 > 1, \qquad \kappa_1(Y = 64, h = 0.03) = 1.7791 > 1.$$

   Because $\cosh \kappa_1$ and $\sinh \kappa_1$ are no longer in the small-hY Taylor regime at $h \in \{0.02, 0.03\}$ when $Y = 64$, the h-fit spread on R_{12} explodes by 3800× between $Y = 4$ (3.11×10^{-8}) and $Y = 64$ (1.19×10^{-4}), and the discrepancy between the finite-sea extrapolation and the exact Wiener–Hopf value grows from 3.26×10^{-9} at $Y = 4$ to 4.33×10^{-5} at $Y = 64$. Thus, for large Y, h must be scaled so that $hY \ll b$ rather than held on an Y-independent h-grid.

2. **37.23× inflation of `mesh_error` by the ill-conditioned `hlog7` basis:** In `twobody_toda_analysis.py` (lines 64–70), `mesh_error` is defined as $\max_{\text{fit}}(\|w_{\text{fit}}\|_1 \max_i(|\Delta y_i| + 2 \times 10^{-10}))$. Because $h^2 \log h$ and $h^3 \log h$ are nearly collinear with polynomials on $h \in [0.001, 0.01]$, the `hlog7` intercept weights have ℓ^1 norm $\|w_{\text{hlog7}}\|_1 = 37.2325$ (versus 3.79 for `quartic7`). Multiplying the hardcoded 2×10^{-10} floor by 37.2325 creates a minimum `mesh_error` of 7.45×10^{-9} everywhere; for the sound speed, the reported `mesh_error` (8.77×10^{-9}) is 39× larger than the actual h-fit spread (2.24×10^{-10}).

3. **Understated $Y \to \infty$ truncation error in $S^{(1)}(r)$ and $\vartheta_N^{(1)}(r)$ at $r = 0.25$:** At $r = 0.25$ (Table 2 of `twobody_toda.tex`), the package's 3-point quadratic fit on $Y \in \{32, 128, 512\}$ achieves an error of 3.18×10^{-8} on $R_{12}^{(1)}(0.25)$, well inside its budget of 1.79×10^{-7}. Yet on $S^{(1)}(0.25)$ and $\vartheta_N^{(1)}(0.25)$, the error jumps by a factor of **2684** to 8.53×10^{-5}, exceeding the unweighted `fit_spread` (3.09×10^{-5}) by 2.76× and consuming 66.3% of the 3×-inflated

budget (1.29×10^{-4}). Equation (15) explains why: $S(Y, r)$ expands in inverse powers of the *smaller* soliton parameter $y_2 = r^2 Y$, with $O(r^{-3}Y^{-1}) = 64/Y$ and $O(r^{-5}Y^{-2}) = 1024/Y^2$ coefficients. At $r = 0.25$ and $Y = 32$, $y_2 = r^2 Y = 2.0 = O(1)$, so $1/y_2 = 0.5$ is not small! As shown in the last column of Table 3, using the exact one-body value $S^{(1)}(0.25) = \frac{17}{4\pi} - \pi$ reduces the error on $\vartheta_N^{(1)}(0.25)$ by 2683× (from 8.52×10^{-5} to 3.18×10^{-8}).

6 Vanishing of the Leading Head-On Directional Coefficient (Point 10)

In Section 3.3, the package studies a right-moving hole $p_1 = q + hu_1$ and a left-moving hole $p_2 = -q - hu_2$. We audit both the mathematical statement for R_{12}^{ho} **(c)** and its physical interpretation **(d)**:

1. **Confirmation of $R_{12}^{\mathrm{ho},(0)}(y_1, y_2) = 0$ (c):** For a hole at $-q-hu_2$, the source $\Theta(x-p_2)-\Theta(x+q)$ in the finite-sea Bethe equation is smooth across the opposite edge $x \approx +q$, where its derivative is $K(2q+O(h))\, hu_2 = O(h)$. Consequently, the dressing backflow near $+q$ is $f_2^+(v) = O(h)$, so both the directional phase N_{12}^{ho}/h and the endpoint reference U_{12}^{ho}/h vanish at $O(h^0)$ ($N_{12}^{\mathrm{ho}}/h = O(h)$, $U_{12}^{\mathrm{ho}}/h = O(h)$). Meanwhile, on the classical side with $\varepsilon_1 = +1, \varepsilon_2 = -1$,

$$\frac{\delta_T^{\mathrm{ho}}(\kappa_1, \kappa_2)}{h} = +4\sqrt{y_1 y_2} + O(h), \qquad \frac{C_{12}^{\mathrm{ho}}(\kappa_1, \kappa_2)}{h} = -4\sqrt{y_1 y_2} + O(h),$$

because $P_{\mathrm{good}}(\kappa_2) = O(\kappa_2^3)$ while $M_{\mathrm{good}}(\kappa_2) = 2b \sinh \kappa_2 = 2\sqrt{bhy_2} + O(h^{3/2})$. The $O(h^0)$ classical contributions $+4\sqrt{y_1 y_2}$ and $-4\sqrt{y_1 y_2}$ cancel identically, leaving $(\delta_T^{\mathrm{ho}} + C_{12}^{\mathrm{ho}})/h = \frac{4\mathrm{e}^{1/2}}{3}(y_1^{3/2} y_2^{1/2} - y_1^{1/2} y_2^{3/2})h + O(h^2) = O(h)$. Thus $R_{12}^{\mathrm{ho}}(y_1, y_2; h) = O(h) \to 0$ as $h \to 0$. In Check 5 of `audit_twobody_toda.py`, our independent 5-point Fejér-I Neville extrapolation on $h \in \{0.064, 0.032, 0.016, 0.008, 0.004\}$ confirms $R_{12}^{\mathrm{ho}} \to 0$, $R_{21}^{\mathrm{ho}} \to 0$, $N_{12}^{\mathrm{ho}}/h \to 0$, and $U_{12}^{\mathrm{ho}}/h \to 0$ to $< 2 \times 10^{-6}$ across all tested pairs $(Y, r) \in \{(4, 0.5), (16, 0.4), (16, 0.5), (16, 0.6)\}$.

2. **Qualification regarding the endpoint-subtracted observable $\mathcal{B}_{\mathrm{ho}}/h$ and physical head-on scattering (d):** In the overtaking channel, the two-term antisymmetric residual $\frac{1}{2}(N_{12} - N_{21})/h$ and the four-term algebraically antisymmetric representative $\mathcal{B}_{12}/h = (F_{12} - M_1 Q_2)/h$ differ only by the one-body edge terms $-\sqrt{\pi}[B(u_1) - B(u_2)]$. In the head-on channel, however, $F_{12}^{\mathrm{ho}}/h = O(h)$, so $\mathcal{B}_{\mathrm{ho}}/h$ is dominated entirely by the endpoint product $-M(p_1)Q(p_2)/h$:

$$\lim_{h\to 0} \frac{\mathcal{B}_{\mathrm{ho}}(p_1, p_2)}{h} = \big(B(u_1) - \sqrt{\pi}\big)\big(B(u_2) - \sqrt{\pi}\big) = 4rY - 2\sqrt{\pi}\sqrt{Y}(1+r) + \pi + \frac{2}{\pi}\left(r + \frac{1}{r}\right) + O(Y^{-1/2}), \tag{17}$$

verified to $< 4 \times 10^{-6}$ by our Fejér-I solver in Check 5. After subtracting the classical benchmark $+4rY$, $\mathcal{B}_{\mathrm{ho}}/h - 4rY$ does *not* vanish, but diverges as $-2\sqrt{\pi}\sqrt{Y}(1 + r)$ and has a finite $O(Y^0)$ piece $\pi + \frac{2}{\pi}(r + r^{-1})$. Physically, a right-moving and a left-moving excitation have relative velocity $\Delta v \approx 2$ rather than $O(h)$, so their collision probes the bulk of the Fermi sea across $[-q, q]$ rather than a common spectral edge. The vanishing of $R_{12}^{\mathrm{ho},(0)}$ is therefore a consequence of the opposite-edge decoupling $K(2q) = O(1)$ and the cancellation $\delta_T^{\mathrm{ho}} + C_{12}^{\mathrm{ho}} = O(h^2)$, not a statement that the physical head-on Boussinesq quantum phase shift is trivial.

7 Failures Recorded

In accordance with `AGENTS.md`, we record all package issues identified by the audit as well as all intermediate failures encountered and fixed while developing `phase2/scripts/audit_twobody_toda.py`:

1. **Package issue 1 (Confirmed error in prior audit `audit_frame_dictionary.tex`):** Equation (9) (line 431) of `phase2/audit_frame_dictionary.tex` and line 797 of `audit_frame_dictionary` printed the four-term Krein characteristic source as $2f(u_1)[B(u_2)-\sqrt{\pi}]$ instead of $2f(u_1)B(u_2)-2\sqrt{\pi}f(u_2)$. As proved in Section 3, the exclusion of $A_\lambda(y_1, y_2)$ in `audit_frame_dictionary.tex` nevertheless holds without modification.

2. **Package issue 2 (Test-coverage gaps in `twobody_toda_*.py`):** Equation (16) (`eq:regular`) of `twobody_toda.tex` was not tested in any package script, and `twobody_toda_identities.py` tested only δ_T/h rather than $(\delta_T + C_{12})/h$. Both are verified in `audit_twobody_toda.py`.

3. **Package issue 3 (Error-budget qualifications at $Y = 64$ and $r = 0.25$):** At $Y = 64$, `quartic7` includes $h = 0.02$ ($\kappa_1 = 1.453 > 1$), and at $r = 0.25$, $Y = 32$ corresponds to $y_2 = r^2 Y = 2.0$, producing an 8.52×10^{-5} truncation error in $S^{(1)}(0.25)$ and $\vartheta_N^{(1)}(0.25)$ that exceeds the unweighted `fit_spread` by 2.76×.

4. **Audit script development failure 1 (`mpmath.mp.dps` mutation across `runpy.run_path`):** In our first run of `audit_twobody_toda.py`, executing `twobody_toda_edge.py` (`mp.dps=32`) and `twobody_toda_analysis.py` (`mp.dps=60`) via `runpy.run_path` in Check 1 mutated the shared global precision `mp.mp.dps` from 70 down to 60, causing the 70-digit algebraic identity check in Check 2 (`assert max(alg_errs) < 1e-65`) to fail at $\sim 10^{-59}$. Restoring `mp.mp.dps = 70` in the `finally` block of Check 1 fixed the issue (max error 5.66×10^{-73}).

5. **Audit script development failure 2 (Check 4 Eq. (16) test and `edge.json` grid lookup, and Check 5 unsubtracted head-on tolerance at $Y = 16$):** In intermediate runs of Checks 4–5, we (a) initially asserted `fwd_disc_025 < 1e-9` before reading the exact value 3.177×10^{-8} from `twobody_toda_grid.json`; (b) used the large-y asymptotic formula as the initial Newton guess in `solve_u_B(z)` on $z \in (0, 1)$ and queried $(Y, r) = (16, 0.25)$ in `twobody_toda_edge.json` (where $r = 0.25$ starts at $Y = 32$); and (c) applied the $Y = 4$ threshold `2e-6` to the unsubtracted head-on N_{12}^{ho}/h at $Y = 16$ on $h \in [0.004, 0.064]$, where $\kappa_1(16, 0.064) = 1.30 > 1$ leaves an $O((hY)^5) \sim 5 \times 10^{-5}$ classical κ-series residual in N_{12}^{ho}/h that cancels out in R_{12}^{ho}. Using the small-y guess $u_0 = \frac{4}{3\sqrt{\pi}} y^{3/2}$ for $y < 0.8$, querying $(32, 0.25)$, and setting `tol_nu = 2e-6` at $Y = 4$ and `2e-4` at $Y = 16$ resolved these issues.

References

- E. Corrigan and C. Zambon, *Integrable defects in the Boussinesq equations*, `arXiv:2306.03772` (verified in project literature dictionary; $u_{\mathrm{CZ}} = -3\varphi$, $L_{\mathrm{CZ}} = 9L$).

- Internal repository documents verified in full text: `phase2/twobody_toda.tex`, `phase2/audit_phase2i.tex`, `phase2/audit_frame_dictionary.tex`, `phase2/frame_dictionary.tex`, `phase2/audit_toda_dictionar`, `phase2/audit_onebody_scheme.tex`, `phase2/audit_smx.tex`, `phase2/errata.tex`, and `phase2/report.t`